

Day-by-day summary (all tracks)

Day 1 (Sept 5) — Contracts + scaffolding Start with a 30–45 min group sync: lock the exact JSON shape of all 4 endpoints, confirm region (Bay of Bengal) and the 5-variable set. Everyone then builds against that contract independently.

Day 2 (Sept 6) — Real data starts replacing mocks Model and in-situ pipelines go live; intelligence engines still running against mocks; API endpoints come online as real data lands.

Day 3 (Sept 7) — Full integration, no UI yet Everything wires together into one real loop: real model data → real in-situ data → Graph Fusion → Drift Memory → alerts, all testable via curl/Postman. **End-of-day target: the full backend loop runs with zero UI.**

Day 4 (Sept 8) — Frontend blitz + record + rehearse 3 people build the CesiumJS/Three.js visualization (scene, trust overlay, live alert feed, time-scrubber) and wire it straight to the now-real endpoints. The other 3 do final bug-fixes, threshold tuning, a stability pass, and prep the backup demo video + pitch script.

De-risking tip: whoever ends up on the Day 4 frontend team should spend an hour on Day 1 wiring a bare CesiumJS scene to mock data, so Day 4 is "swap in real endpoints and polish," not "build from zero."

Fallback order if time runs short (never cut the core loop)

1. Nowcast Engine — cut first, becomes a "Future Scope" slide bullet
2. Full PyTorch Geometric GNN — fall back to simpler graph-weighted correction
3. Expert/Public mode toggle — fall back to expert view only
4. Multi-variable support — fall back to SST only if currents/waves aren't stable in time
5. **Never cut:** real model data + real in-situ data + at least one visible fusion signal

- **AFTER 6 TRACKS FULL WORK DONE - UI/UX + TESTING**

Frontend / Visualization team (3 people)

Person 1 (you) — UI/UX Lead

- Overall visual design direction — layout, color language, typography, the "wow factor" pass
- Trust overlay design — making the green/red confidence coding actually read clearly on stage, not just function
- Expert/Public mode toggle — design + interaction (fallback: expert-only if time runs short)
- Final integration pass tying the whole scene together before recording

Person 4 — Data Wiring

- Wires real `/model` and `/fused` output into the CesiumJS/Three.js scene (ocean surface, currents, isosurface/marching cubes) — makes sense since you already know this data's shape and units from Days 1–3
- Colorbar editor + vertical exaggeration slider

Person 5 — Live Data Wiring

- Wires WebSocket alerts into the live alert feed panel — you already own this pipe
- Time-scrubber, wired to real forecast window data
- Click-to-inspect profile chart, pulling from `/observations` and `/model`

Stability & Testing team (3 people)

Person 2

- Threshold tuning on Drift Memory using real Day 3 integration data
- Confirm `/alerts` fires correctly under the actual live-demo scenario

Person 3

- Go/no-go call on Nowcast Engine based on Day 3 status — cut cleanly to "Future Scope" slide if not stable
- Bug fixes on the Streaming Backbone

Person 6

- End-to-end smoke test of the full loop, focused on in-situ data edge cases (missing readings, timestamp mismatches) — the kind of thing that quietly breaks a live demo